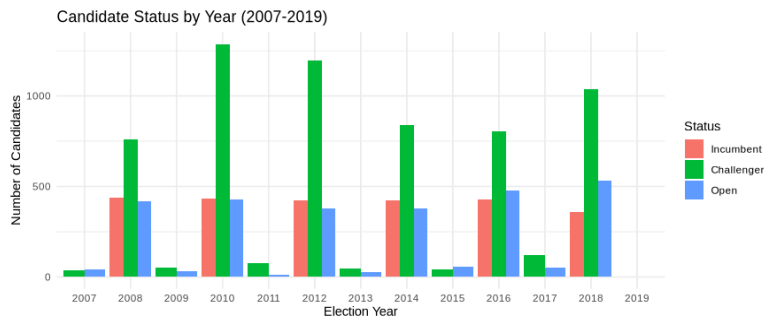
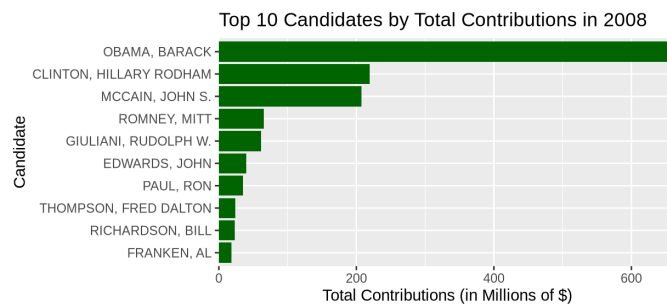


Part 1



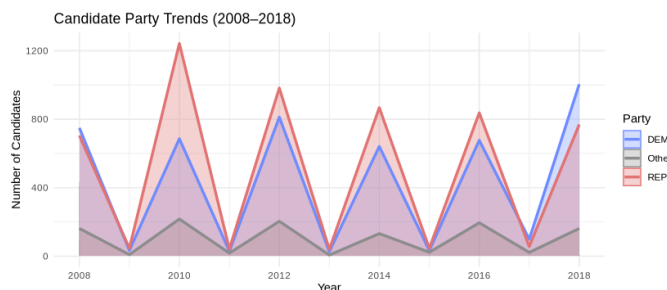
This plot allows us to compare the number of Incumbents, Challengers, and Open seat candidates across multiple election cycles. We can see in the odd years the off-year election. Challengers usually dominate the candidates in each years especially even years. We can also see the biggest spikes in challenger candidates were in 2010 and 2012, suggesting a

potential shift in the House or Senate.



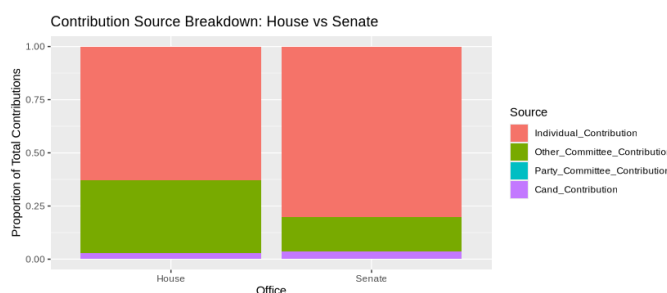
This horizontal bar chart highlights the top 10 fundraisers in the 2008 election cycle. It shows who attracted the most financial backing and allows comparison between presidential and congressional candidates. Most of the top fundraisers were Presidential candidates, reflecting the high fundraising demands of national campaigns. The steep drop after the top 3 suggests a strong concentration of donor funding among leading contenders.

3 suggests a strong concentration of donor funding among leading contenders.



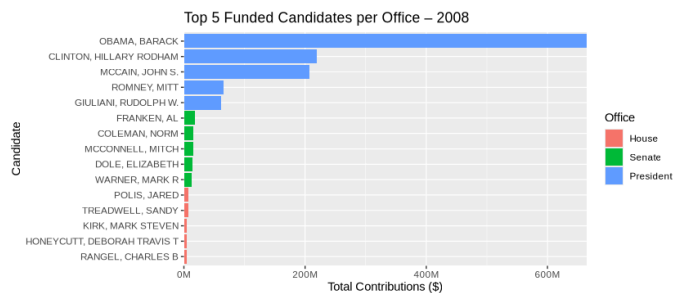
This graph illustrates the trends in the number of candidates affiliated with the Democratic, Republican, and other parties from 2008 to 2018. Across all years, the Republican Party consistently fielded more candidates than the Democratic Party, with noticeable spikes during midterm election years (2010, 2014). The "Other" category remains relatively small and

stable throughout, highlighting the continued dominance of the two major parties in U.S. elections.



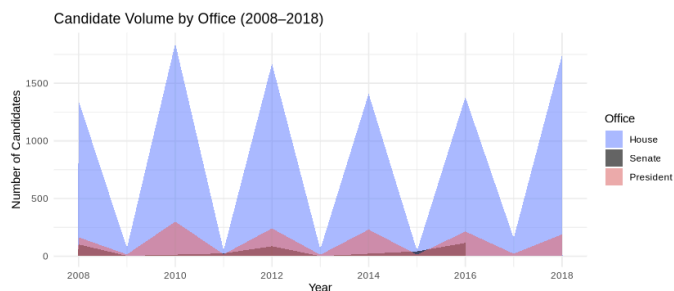
The graph shows the breakdown of campaign funding sources for House and Senate candidates. For both chambers, the majority of funding comes from individual contributions, with House candidates relying slightly more on other

committee contributions than Senate candidates. Contributions from party committees make up a much smaller portion overall, but are still present.



This graph displays the top 5 highest-funded candidates from the House, Senate, and Presidential races in the 2008 election cycle. A clear pattern emerges: presidential candidates significantly outpaced Senate and House candidates in total contributions. Even the top Senate and House fundraisers appear modest in comparison, revealing the vastly

greater financial demands and fundraising capacity associated with national presidential campaigns.

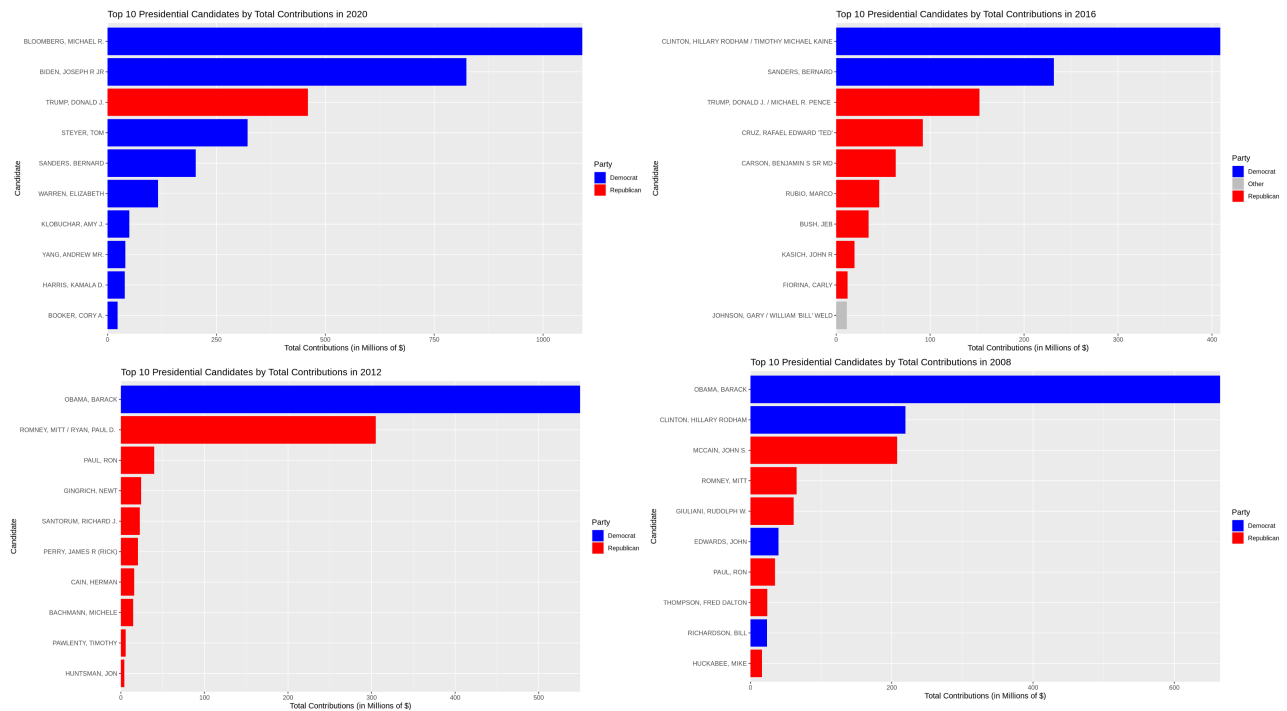


This area plot shows the number of House, Senate, and Presidential candidates from 2008 to 2018. The most noticeable trend is the cyclical spike in House candidates during midterm years, with lower participation during presidential election years. Senate and Presidential races show a more stable, lower volume

of candidates overall, with slight increases in presidential years. This pattern reflects the fixed 2-year term for House seats, leading to frequent elections and high turnover.

Part 2

Hypothesis 1:



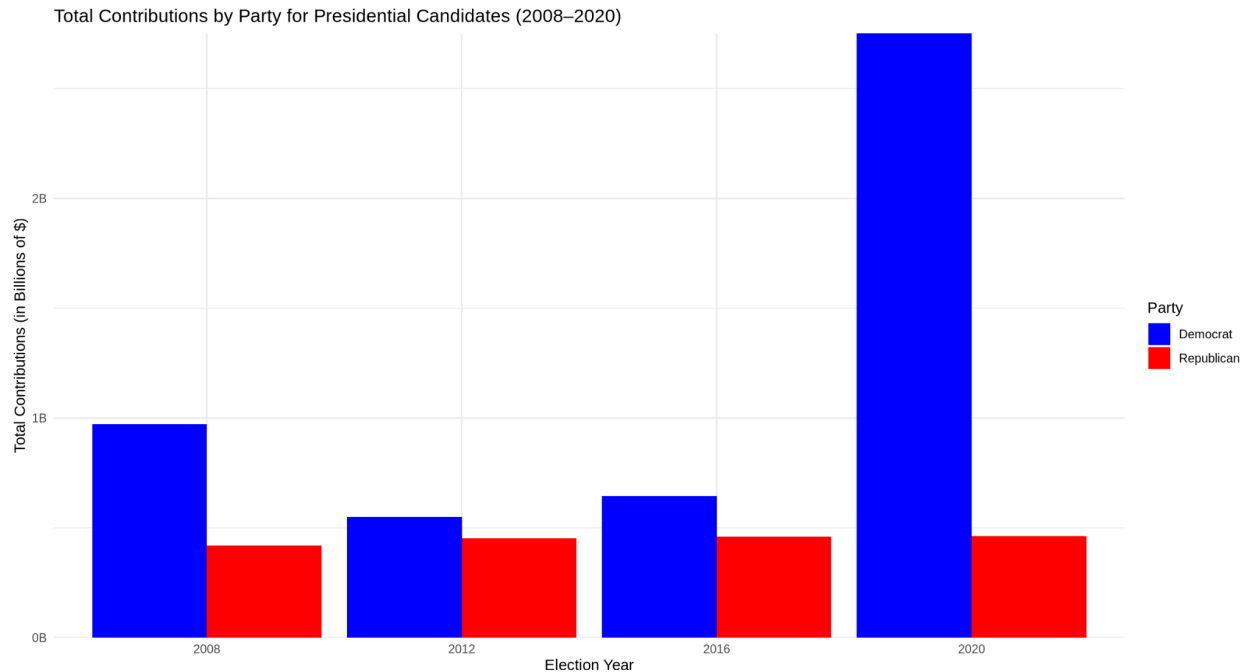
Hypothesis 1: The more funds a candidate raises, the higher the probability they are likely to be elected as their party's nominee

We plotted out the top 10 presidential candidates by total contribution for each of the election years (2008, 2012, 2016, & 2020.) In three of the four election years ('08, '12, '16), we can see the hypothesis stand, as the top for each party is then nominated.

There is one outlier in the 2020 dataset, Democratic Presidential Campaign dataset. Michael Bloomberg had raised the most funds, but was mostly raised by his own wealth. Also, he entered the race late and ended the campaign early to endorse Joe Biden. Therefore, Joe Biden became the top fundraiser among the Democratic candidates after Michael Bloomberg stepped down, and thus Joe Biden became the Democratic nominee for president.

Self-funding candidates (e.g., Bloomberg, Steyer) often raise lots of money but lack broad voter appeal or party support. Which does show that massive fundraising does not guarantee a nomination if other political dynamics override. The graphs largely supports the hypothesis that higher fundraising correlates with a greater likelihood of becoming the party nominee, especially when the fundraising is paired with electoral viability. However, fundraising alone is not sufficient, and exceptions do exist, typically influenced by factors like political experience, media attention, voter enthusiasm, and party alignment.

Hypothesis 2: Democrat Presidential Candidates raise more money than Republican candidates to fund their campaigns



We plotted out the total contribution for the Democrat and Republican Party for each of the election years (2008, 2012, 2016 & 2020.) For every presidential election year we can see the hypothesis that Democrat candidates have raised noticeably more funds compared to the Republican candidates.

During the Obama administration, when the Democrats had held the presidency ('12, '16), Democrat candidates still raised much more funds compared to the Republicans. However, there is no significant difference between the two parties. On the other hand, under the Bush administration (2008) and the Trump administration (2020) the total contribution difference had differed over double.

A possible explanation is that in years like 2020 and 2008, they featured a large field of high-profile candidates (e.g., Obama, Clinton, Sanders, Biden), driving up total donations. We also see that the contributions for democrats fluctuate throughout the years, while for the republicans it is a stable amount each time.

Code

```
library(ggplot2)
library(scales)
library(dplyr)

a2data <- read.csv("IN4MTX 143/fec_2008-2022 (1).csv")

#####

#####

## Plot 1
# a2data$Year <- as.numeric(format(as.Date(a2data$Coverage_End_Date, "%m/%d/%Y"),
"%Y"))
# data_pre2020 <- subset(a2data, Year <= 2019)
#
# data_pre2020$Cand_Incumbent_Challenger_Open_Seat <-
toupper(trimws(data_pre2020$Cand_Incumbent_Challenger_Open_Seat))
#
# valid_statuses <- c("INCUMBENT", "CHALLENGER", "OPEN")
# data_pre2020 <- subset(data_pre2020, Cand_Incumbent_Challenger_Open_Seat %in%
valid_statuses)
#
# data_pre2020$Status <- factor(data_pre2020$Cand_Incumbent_Challenger_Open_Seat,
#                               levels = c("INCUMBENT", "CHALLENGER", "OPEN"),
#                               labels = c("Incumbent", "Challenger", "Open"))
#
# ggplot(data_pre2020, aes(x = as.factor(Year), fill = Status)) +
#   geom_bar(position = "dodge") +
#   labs(title = "Candidate Status by Year (2007-2019)",
#        x = "Election Year", y = "Number of Candidates", fill = "Status") +
#   theme_minimal()

#####

#####

## Plot 2
# data_2008 <- subset(a2data, format(as.Date(Coverage_End_Date, "%m/%d/%Y"), "%Y") ==
"2008")
#
# top10_2008 <- head(data_2008[order(-data_2008$Total_Contribution), ], 10)
#
# top10_2008$Total_Contribution_Millions <- top10_2008$Total_Contribution / 1e6
#
```

```

# ggplot(top10_2008, aes(x = reorder(Cand_Name, Total_Contribution_Millions), y =
Total_Contribution_Millions)) +
# geom_bar(stat = "identity", fill = "darkgreen") +
# coord_flip() +
# scale_y_continuous(expand = c(0, 0)) +
# labs(title = "Top 10 Candidates by Total Contributions in 2008",
#       x = "Candidate",
#       y = "Total Contributions (in Millions of $)")

```

```

#####
#####

```

```

## Plot 3
# subset_years <- subset(a2data, format(as.Date(Coverage_End_Date, "%m/%d/%Y"), "%Y")
%in% as.character(2008:2018))
# subset_years$Year <- as.numeric(format(as.Date(subset_years$Coverage_End_Date,
"%m/%d/%Y"), "%Y"))
#
# subset_years$Simplified_Party <- ifelse(subset_years$Cand_Party_Affiliation %in% c("DEM",
"REP"),
#                                       subset_years$Cand_Party_Affiliation, "Other")
#
# party_year_counts <- aggregate(Cand_Id ~ Year + Simplified_Party, data = subset_years,
function(x) length(unique(x)))
#
# ggplot(party_year_counts, aes(x = Year, y = Cand_Id, fill = Simplified_Party, color =
Simplified_Party)) +
# geom_area(alpha = 0.3, position = "identity") +
# geom_line(size = 1.2) +
# scale_x_continuous(breaks = seq(2008, 2018, 2)) +
# scale_fill_manual(values = c("DEM" = "#718dff", "REP" = "#e57373", "Other" = "#8d8d8d")) +
# scale_color_manual(values = c("DEM" = "#718dff", "REP" = "#e57373", "Other" =
"#8d8d8d")) +
# labs(title = "Candidate Party Trends (2008–2018)",
#       x = "Year", y = "Number of Candidates", color = "Party", fill = "Party") +
# theme_minimal()

```

```

#####
#####

```

```

## Plot 4
# subset_chambers <- subset(a2data, Cand_Office %in% c("H", "S"))
#
# subset_chambers$Office <- ifelse(subset_chambers$Cand_Office == "H", "House", "Senate")

```

```
#
# stacked_contribs <- subset_chambers[, c("Office", "Individual_Contribution",
"Other_Committee_Contribution", "Party_Committee_Contribution", "Cand_Contribution")]
#
# library(reshape2)
# long_contribs <- melt(stacked_contribs, id.vars = "Office", variable.name = "Source",
value.name = "Amount")
# long_contribs <- aggregate(Amount ~ Office + Source, data = long_contribs, FUN = sum)
#
# ggplot(long_contribs, aes(x = Office, y = Amount, fill = Source)) +
#   geom_bar(stat = "identity", position = "fill") +
#   labs(title = "Contribution Source Breakdown: House vs Senate (2008-2018)",
#         x = "Office", y = "Proportion of Total Contributions", fill = "Source")
#
```

```
#####
#####
```

Plot 5

```
# get_top5_by_office_year <- function(df, year, office) {
#   df$Year <- as.numeric(format(as.Date(df$Coverage_End_Date, "%m/%d/%Y"), "%Y"))
#   year_data <- subset(df, Year == year & Cand_Office == office)
#   year_data <- year_data[order(-year_data$Total_Contribution), ]
#   return(head(year_data, 5))
# }
#
# top_house_2008 <- get_top5_by_office_year(a2data, 2008, "H")
# top_senate_2008 <- get_top5_by_office_year(a2data, 2008, "S")
# top_pres_2008 <- get_top5_by_office_year(a2data, 2008, "P")
#
# library(dplyr)
# top_all_2008 <- bind_rows(top_house_2008, top_senate_2008, top_pres_2008)
# top_all_2008$Cand_Office <- factor(top_all_2008$Cand_Office, levels = c("H", "S", "P"),
labels = c("House", "Senate", "President"))
#
# ggplot(top_all_2008, aes(x = reorder(Cand_Name, Total_Contribution), y = Total_Contribution,
fill = Cand_Office)) +
#   geom_bar(stat = "identity") +
#   coord_flip() +
#   scale_y_continuous(
#     labels = label_number(scale = 1e-6, suffix = "M"),
#     expand = c(0, 0)
#   ) +
```

```
# labs(title = "Top 5 Funded Candidates per Office – 2008",  
#       x = "Candidate", y = "Total Contributions ($)", fill = "Office")
```

```
#####
```

```
## Plot 6
```

```
a2data$Year <- as.numeric(format(as.Date(a2data$Coverage_End_Date, "%m/%d/%Y"),  
"%Y"))
```

```
# Filter years and offices
```

```
count_by_office <- a2data %>%  
  filter(Cand_Office %in% c("H", "S", "P"), Year >= 2008 & Year <= 2018) %>%  
  group_by(Year, Cand_Office) %>%  
  summarise(Count = n_distinct(Cand_Id))
```

```
# Plot
```

```
ggplot(count_by_office, aes(x = Year, y = Count, fill = Cand_Office)) +  
  geom_area(alpha = 0.6, position = "identity") + # Lower alpha for more transparency  
  scale_fill_manual(values = c("H" = "#718dff", "S" = "#e57373", "P" = "black"),  
                    labels = c("House", "Senate", "President")) +  
  scale_x_continuous(breaks = seq(2008, 2018, 2)) +  
  labs(title = "Candidate Volume by Office (2008–2018)",  
       x = "Year", y = "Number of Candidates", fill = "Office") +  
  theme_minimal()
```

```
#####
```

```
## Part 2 Code
```

```
## Hypothesis 1
```

```
data_2008 <- subset(a2data, format(as.Date(Coverage_End_Date, "%m/%d/%Y"), "%Y")  
  == "2008" & Cand_Office == "P")
```

```
top10_2008 <- head(data_2008[order(-data_2008$Total_Contribution), ], 10)
```

```
top10_2008$Total_Contribution_Millions <- top10_2008$Total_Contribution / 1e6
```

```
top10_2008$Party_Color <- ifelse(top10_2008$Cand_Party_Affiliation == "DEM", "Democrat",  
  ifelse(top10_2008$Cand_Party_Affiliation == "REP", "Republican",  
  "Other"))
```

```
ggplot(top10_2008, aes(x = reorder(Cand_Name, Total_Contribution_Millions),  
  y = Total_Contribution_Millions,
```

```

    fill = Party_Color)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  scale_fill_manual(values = c("Democrat" = "blue", "Republican" = "red", "Other" = "gray")) +
  scale_y_continuous(expand = c(0, 0)) +
  labs(title = "Top 10 Presidential Candidates by Total Contributions in 2008",
    x = "Candidate",
    y = "Total Contributions (in Millions of $)",
    fill = "Party")

```

Hypothesis 2

```

pres_data <- subset(a2data, Cand_Office == "P" &
  format(as.Date(Coverage_End_Date, "%m/%d/%Y"), "%Y") %in%
  c("2008", "2012", "2016", "2020"))

pres_data$Year <- as.numeric(format(as.Date(pres_data$Coverage_End_Date, "%m/%d/%Y"),
"%Y"))
pres_data$Party <- ifelse(pres_data$Cand_Party_Affiliation == "DEM", "Democrat",
  ifelse(pres_data$Cand_Party_Affiliation == "REP", "Republican", NA))

pres_data <- subset(pres_data, Party %in% c("Democrat", "Republican"))

summary_data <- pres_data %>%
  group_by(Year, Party) %>%
  summarise(Total_Contribution = sum(Total_Contribution, na.rm = TRUE))

ggplot(summary_data, aes(x = factor(Year), y = Total_Contribution, fill = Party)) +
  geom_bar(stat = "identity", position = "dodge") +
  scale_fill_manual(values = c("Democrat" = "blue", "Republican" = "red")) +
  scale_y_continuous(
    labels = label_number(scale = 1e-9, suffix = "B"),
    expand = c(0, 0)
  ) +
  labs(title = "Total Contributions by Party for Presidential Candidates (2008–2020)",
    x = "Election Year",
    y = "Total Contributions (in Billions of $)",
    fill = "Party") +
  theme_minimal()

```

Citation

Merica, D. (2020, March 4). *Michael Bloomberg ends 2020 presidential campaign* | CNN politics. CNN. <https://www.cnn.com/2020/03/04/politics/michael-bloomberg-dropping-out>

Mollie. (2013, August 22). *Date formats in R: R-bloggers*. R. https://www.r-bloggers.com/2013/08/date-formats-in-r/#google_vignette

Subset - make a table showing the 10 largest values of a variable in R? - stack overflow. (n.d.). <https://stackoverflow.com/questions/31939643/make-a-table-showing-the-10-largest-values-of-a-variable-in-r>

Station, D. (2023, May 15). *[Data Analysis with R Basics] 21. Handling Date Data Preprocessing / Tidyverse / Lubridate*. YouTube. <https://www.youtube.com/watch?v=KLzebqyJgjE>